

EldoGas

On-Demand LPG Delivery App

Product Concept & User Flow Document — Complete Edition

Gas delivered to your doorstep — fast, safe, and reliable.

Single-shop model | Swap & New Cylinder | Pay on Delivery

This document is the authoritative, combined product specification integrating the original concept document with all supplementary additions and gap resolutions.

1. Business Overview

EldoGas is a mobile-first, on-demand LPG delivery platform serving households, restaurants, and businesses across Kenyan urban and peri-urban areas. The model is built around speed, safety, and simplicity — a single shop operation dispatching verified riders to deliver pre-filled cylinders directly to customers' doors.

1.1 How the Business Works

The shop holds stock of pre-filled cylinders in all available sizes. When a customer places an order through the app, the nearest available rider is assigned, picks up the correct pre-filled cylinder from the shop, rides to the customer, completes the swap or new cylinder handover, collects payment on the spot, and returns the empty cylinder to the shop for refilling. The cycle is clean, fast, and requires no third-party vendors.

1.2 Business Model at a Glance

Parameter	Detail
Operation Type	Single shop, multiple riders
Delivery Model	Swap (empty for full) + New Cylinder
Payment	Pay on delivery — Cash or M-Pesa Till
Target Market	Households, restaurants, offices, institutions
Cylinder Sizes	3kg, 6kg, 13kg, 25kg, 50kg
Key Differentiator	Speed, safety certification, and doorstep convenience

Admin-managed pricing — important note

All prices shown throughout this document are placeholder/dummy values used for illustration only.

The Admin is the sole authority for configuring prices, cylinder sizes, gas brands, and add-on accessories

via the Admin dashboard. Nothing is hard-coded into the app. See Section 10 for full details.

2. Order Types

EldoGas offers two clearly defined order types. Every customer interaction flows through one of these two paths.

2.1 Swap Order

The core, recurring order type. The customer already has a cylinder at home — they just need it refilled. The rider arrives with a pre-filled cylinder, takes the empty one, and collects payment. From the customer's perspective it is effortless: they do not carry anything, travel anywhere, or wait in queues.

Swap Order Flow

1. Customer has empty cylinder at home
2. Places swap order on the app (selects size and brand)
3. Rider arrives with a pre-filled cylinder
4. Customer hands over empty cylinder
5. Customer pays on delivery (Cash or M-Pesa Till)
6. Rider returns empty cylinder to shop

2.2 New Cylinder Order

For first-time buyers or customers who need an additional cylinder. The customer orders a brand new cylinder that arrives already filled and ready to use. After this first purchase, they automatically enter the swap loop (saving the customer gas brand for easy ordering) for all future orders. The new cylinder order screen also allows customers to add accessories depending on the cylinder size selected.

New Cylinder Order Flow

1. Customer selects 'I need a new cylinder'
2. Selects cylinder size: 3kg / 6kg / 13kg / 25kg / 50kg
3. Selects optional accessories based on size (see Section 4)
4. Reviews total price (Cylinder Price + Gas + Accessories + Free Delivery)
5. Rider delivers brand new filled cylinder plus any accessories
6. Customer pays full amount on delivery
7. Customer is now in the swap loop for future orders

3. Pricing Structure

All pricing is displayed transparently in the app before the customer confirms their order. The delivery fee is shown separately from the gas/cylinder cost. For new cylinder orders, the Cylinder Price is also broken out as its own line item so customers understand they are paying for a reusable asset, not a one-time purchase.

All prices are placeholder values — managed by the Admin

Every price in this section is a dummy/placeholder figure. The Admin sets and updates all prices via the Admin dashboard (Section 10). Price changes take effect immediately across the live app.

3.1 Swap Order Pricing

Cylinder Size	Gas Refill Price (Placeholder)	Delivery Fee (Placeholder)
3kg	Ksh 350	Ksh 100
6kg	Ksh 900	Ksh 100
13kg	Ksh 1,850	Ksh 100
25kg	Ksh 3,500	Ksh 250 — commercial rate
50kg	Ksh 6,500	Ksh 400 — commercial rate

3.2 New Cylinder Pricing (Cylinder Price + Gas Fill)

Cylinder Size	Cylinder Price (Placeholder)	Gas Fill Price (Placeholder)
3kg	Ksh 1,500	Ksh 350
6kg	Ksh 2,500	Ksh 900
13kg	Ksh 3,500	Ksh 1,850
25kg	Ksh 9,000	Ksh 3,500
50kg	Ksh 18,000	Ksh 6,500

Commercial delivery fee

25kg and 50kg cylinders require a physically stronger rider and possible use of a hand trolley. The Admin can set a separate delivery fee tier for commercial sizes. This is configured per size in the Admin dashboard under Delivery Fees.

4. New Cylinder Add-Ons by Size

When a customer selects a new cylinder, the app offers relevant accessories based on the cylinder size chosen. Add-ons are optional and the price summary updates in real time as items are selected or deselected. All add-on prices are placeholder values set by the Admin.

4.1 3kg Cylinder — Add-Ons (Multi-Select)

The 3kg cylinder is a simple household size. Customers can select one, multiple, or no add-ons.

Add-On Item	Description	Price (Placeholder)
Burner / Single Burner Stove	Tabletop burner for everyday cooking	+ Ksh 850
Portable Gas Grill	Compact grill, ideal for small cylinders	+ Ksh 2,500

4.2 6kg Cylinder — Add-Ons (Multi-Select)

The 6kg cylinder is popular with small households and light restaurant use. Customers can select one, multiple, or no add-ons.

Add-On Item	Description	Price (Placeholder)
Burner / Single Burner Stove	Tabletop burner for everyday cooking	+ Ksh 850
Gas Pipe / Hose	Hose pipe for 6kg regulators	+ Ksh 250
Regulator	Standard low-pressure regulator — 6kg	+ Ksh 350
Portable Gas Grill	Compact grill, ideal for 6kg cylinder	+ Ksh 2,500

UX Note — 6kg Add-Ons

Use ADD toggle buttons (not radio buttons) so customers can select multiple items.

Show a product photo next to each item — many customers do not know what a regulator looks like.

Display a tip: 'Complete starter set = Burner + Regulator + pre-filled cylinder' to guide first-timers.

4.3 13kg Cylinder — Add-Ons (Single-Select per Category)

The 13kg cylinder is the household and restaurant workhorse. Radio buttons are used within each category because a customer only needs one regulator type and one hose length.

Regulator Options (select one)

Option	Best For	Price (Placeholder)
None	Customer already has a regulator	Ksh 0
Standard Regulator	For standard gas cookers	+ Ksh 400
Heavy Duty Regulator	For large cookers and restaurant use	+ Ksh 650
Table Top Burner	Ceramic infrared plate, fast heating	+ Ksh 500

Gas Pipe / Hose Options (select one)

Option	Best For	Price (Placeholder)
None	Customer already has a hose	Ksh 0
1.5m Hose	Standard household cooker	+ Ksh 250
2.5m Hose	Cooker positioned further from cylinder	+ Ksh 380

UX Note — 13kg Add-Ons

Display a helpful tip: 'Most cookers need a Heavy Duty Regulator + 1.5m Hose. Not sure? Call us before ordering.'

Include a visible [Call Shop] button on this screen to recover customers who are uncertain.

Show product photos for each regulator type so customers can visually identify what they already have at home.

4.4 25kg Cylinder — Add-Ons (Single-Select per Category)

The 25kg cylinder serves small restaurants, cafes, and offices. Add-ons focus on commercial-grade components.

Regulator Options (select one)

Option	Best For	Price (Placeholder)
None	Customer already has a regulator	Ksh 0
Standard Regulator	Light commercial / single-burner setups	+ Ksh 400
Heavy Duty Regulator	Commercial kitchens, large cookers	+ Ksh 650
High-Pressure Regulator	Industrial burners and specialised equipment	+ Ksh 950

Gas Hose / Pipe Options (select one)

Option	Best For	Price (Placeholder)
None	Customer already has a hose	Ksh 0
1.5m Hose	Cooker placed close to the cylinder	+ Ksh 250
2.5m Hose	Cooker positioned further from cylinder	+ Ksh 380
5m Hose	Fixed kitchen installation, commercial use	+ Ksh 600

Cylinder Stand / Trolley (select one)

Option	Best For	Price (Placeholder)
None	Customer already has a stand or space	Ksh 0
Cylinder Stand	Keeps 25kg cylinder stable and upright indoors	+ Ksh 800
Cylinder Trolley	Allows safe movement of cylinder within premises	+ Ksh 1,500

UX Note — 25kg Add-Ons

Show product photos for each regulator and hose type.

Display a tip: 'Commercial starter set = Heavy Duty Regulator + 2.5m Hose.'

Include a [Call Shop] button — commercial customers often want to confirm before ordering.

4.5 50kg Cylinder — Add-Ons (Single-Select per Category)

The 50kg cylinder is for high-volume commercial use — large restaurants, hotels, schools, and institutions. Multi-cylinder manifold connectors are offered for customers running two or more cylinders in parallel.

Regulator Options (select one)

Option	Best For	Price (Placeholder)
None	Customer already has a regulator	Ksh 0
Heavy Duty Regulator	Standard commercial kitchen use	+ Ksh 650
High-Pressure Regulator	Industrial burners, large-scale cooking setups	+ Ksh 950
Dual-Stage Regulator	Consistent pressure for sensitive equipment	+ Ksh 1,400

Gas Hose / Pipe Options (select one)

Option	Best For	Price (Placeholder)
None	Customer already has a hose	Ksh 0
2.5m Hose	Standard commercial kitchen	+ Ksh 380
5m Hose	Cylinder stored away from cooking area	+ Ksh 600
10m Hose	Cylinder in separate room or outdoor storage	+ Ksh 950

Mobility & Installation (select one)

Option	Best For	Price (Placeholder)
None	Customer has existing storage solution	Ksh 0
Heavy-Duty Trolley	Safe movement of 50kg cylinder within premises	+ Ksh 2,200
Cylinder Cage / Bracket	Secure outdoor or wall-mounted cylinder storage	+ Ksh 2,800

Multi-Cylinder Manifold (select one)

Option	Best For	Price (Placeholder)
None	Single cylinder setup	Ksh 0
2-Cylinder Manifold Connector	Two 50kg cylinders in parallel for continuous flow	+ Ksh 3,500

UX Note — 50kg Add-Ons

Include a prominent [Call Shop] button — commercial buyers often want to confirm before committing.

Display a tip: 'Running two cylinders? Add a manifold connector for uninterrupted gas supply.'

Consider an optional advance-booking flow for 50kg orders to allow the shop to prepare logistics.

5. Full Consumer App User Flow

5.1 Onboarding (First-Time User)

The onboarding flow is kept minimal — only the information essential to place a first order is collected upfront. Additional profile details can be filled in later.

Onboarding Steps

1. Download the EldoGas app
2. Enter phone number → Receive OTP → Verify
3. Enter name
4. Set delivery location: use current GPS location (default) or enter name on the map to auto-fill
5. Label the address: Home / Office / Restaurant
6. Land on Home Screen

Why the map pin matters

Many Kenyan addresses are informal or vague (e.g. 'near the blue gate, third left after the petrol station').

The map pin is the real address. Make the pin-drop screen large, full-screen, and easy to zoom and drag.

Also provide a delivery notes field for additional directions.

5.2 Home Screen

The home screen is a dashboard, not a menu. It should communicate everything a returning customer needs at a glance without requiring navigation.

Home Screen Element	Purpose
Shop Status	Open (green dot) or Closed (red dot) — shows opening time if closed
Estimated Delivery Time	'Delivery in ~25 mins to your area-location'
Order Now Button	Large, prominent CTA — central element of the screen
Last Order Summary	'Last swap: 13kg Total — 18 Jan' for one-tap reorder
GasPoints Balance	Shown subtly at the top of the screen
Safety Tip Banner	Rotating safety tip at the bottom (nice-to-have)

5.3 Placing a Swap Order

This is the most frequent user journey — a returning customer placing a routine refill. The flow is optimised for speed.

Swap Order Flow

1. Tap 'Order Gas Now'
2. Screen: 'Do you have an empty cylinder?' → YES → Swap flow / NO → New Cylinder flow
3. Select cylinder size: 3kg / 6kg / 13kg / 25kg / 50kg
4. Select brand (auto-fills from previous order for returning customers — with option to change)
5. Price breakdown displayed: Gas refill price + Delivery fee + Total on delivery
6. Confirm delivery address (pre-filled from profile — can edit for this order)
7. Add delivery notes: e.g. 'Blue gate, 3rd floor, call on arrival'
8. Select payment method: Cash / M-Pesa Till (If M-Pesa Till selected: till number and exact amount shown)
9. Tap 'Place Order'
10. Order confirmation screen shown with order number
11. Live tracking begins automatically

5.4 Placing a New Cylinder Order

For first-time buyers or customers needing an additional cylinder. The flow branches after size selection to show relevant accessories.

New Cylinder Order Flow

1. Tap 'Order Gas Now' → 'I don't have a cylinder / I need a new one'
2. Select cylinder size: 3kg / 6kg / 13kg / 25kg / 50kg
3. If 3kg: add-ons screen shown (Burner, Grill — multi-select)
4. If 6kg: add-ons screen shown (Burner, Regulator, Grill, Hose — multi-select)
5. If 13kg: add-ons screen shown (Regulator type — radio; Hose length — radio)
6. If 25kg: add-ons screen shown (Regulator — radio; Hose — radio; Stand/Trolley — radio)
7. If 50kg: add-ons screen shown (Regulator — radio; Hose — radio; Mobility — radio; Manifold — radio)
8. Live price summary updates as add-ons are toggled (Cylinder Price + Gas Fill + Add-Ons + Delivery)
9. Confirm delivery address
10. Add delivery notes
11. Select payment method: Cash / M-Pesa Till
12. Tap 'Place Order'
13. Confirmation screen and live tracking begin

5.5 Live Tracking Screen

This is the most important screen in the consumer app. Once an order is placed, this screen becomes the entire experience. It should be clear, reassuring, and provide everything the customer needs without requiring them to call the shop or the rider.

Screen Element	Detail
Order Status Bar	4 stages: Order Received → Rider Assigned → On the Way → Delivered
Live Map	Rider's moving dot and customer's pin shown in real time

Screen Element	Detail
Rider Card	Photo, name, star rating, Safety Certified badge, call button
ETA Countdown	'Arriving in ~18 mins' — updates in real time
Order Summary	Collapsible section showing size, brand, and amount to pay
M-Pesa Reference	Till number shown again if M-Pesa was selected
SMS Mirroring	Every status change also sent via SMS for users without data

5.6 On Delivery Handover

Delivery Handover Steps

1. Customer receives notification: 'Your rider is here!'
2. Customer brings out empty cylinder
3. Rider hands over pre-filled cylinder (and any accessories if new cylinder order)
4. Customer pays: Cash handed to rider OR M-Pesa sent to till number
5. Rider marks order as 'Delivered' on their app
6. Customer app shows: 'Delivery Complete' with confirmation checkmark
7. Rating prompt appears automatically

5.7 Post-Delivery Rating Screen

The rating screen is shown immediately after delivery confirmation. It is designed to be quick — three taps and done — while capturing useful feedback data.

Rating Screen Steps

1. 'How was your delivery?' — star rating (1 to 5)
2. Quick-tap tags: Fast / Friendly / Safe handling / Professional
3. Optional: written review field
4. Option: 'Flag an issue' (wrong cylinder, safety concern, rude rider)
5. GasPoints earned displayed: '+100 points for this order!' (or first-order bonus amount)
6. Referral nudge: 'Share your code and earn 250 GasPoints'
7. Done — returns to home screen

6. Rider App Flow

The rider app is a separate, lightweight app. Riders need fast, clear information — not complexity. The design philosophy: minimal steps, instant readability.

6.1 Rider Home Screen

Element	Function
Availability Toggle	Available (green) / Unavailable (red) — one tap to go online or offline
Today's Stats	Orders completed / Earnings / Rating — shown at a glance
Order Notifications	Pop-up card when a new order is assigned

6.2 Incoming Order Notification

When a new order is dispatched, the rider receives a notification card with all the information needed to decide whether to accept. A 50-second countdown timer applies — if the rider does not respond, the order moves to the next available rider.

Sample Incoming Order Card — Standard

NEW ORDER #1042
 Pickup: Your Shop (0.2km away)
 Deliver to: Kilimani, Nairobi [map pin]
 Order: 13kg Total — New Cylinder
 Pack: Heavy Duty Regulator + 1.5m Hose
 Collect on delivery: Ksh 6,350 (M-Pesa)
 Distance: 3.4km | Est. time: 22 mins
 [ACCEPT] [DECLINE]

Sample Incoming Order Card — Commercial (25kg / 50kg)

NEW ORDER #1087 [COMMERCIAL]
 Pickup: Your Shop (0.2km away)
 Deliver to: Westlands, Nairobi [map pin]
 Order: 25kg Swap — Total Gas
 Weight: 25kg | Trolley required: YES
 Collect on delivery: Ksh 3,750 (M-Pesa)
 Distance: 5.1km | Est. time: 28 mins
 [ACCEPT] [DECLINE]

The 'Pack' line and 'COMMERCIAL' flag are critical

For orders with accessories, the rider's card must clearly list every item to be packed from the shop.

This eliminates packing errors and reduces rider return trips.

25kg and 50kg orders show a [COMMERCIAL] flag and trolley requirement so the shop assigns the right rider.

6.3 Active Delivery Flow (Rider)

Active Delivery Steps

1. Accept order
2. Navigate to shop — Google Maps integration
3. At shop: pick up correct pre-filled cylinder (and accessories if applicable)
4. Tap 'Picked Up' to update order status (stock count auto-deducted at this point)
5. Navigate to customer address
6. Complete safety checklist: cylinder condition, valve check, safe handover, empty collected, payment confirmed
7. Tap 'Confirm Delivery'
8. Earnings update in real time on rider's dashboard
9. Return empty cylinder to shop (swap orders)

6.4 Rider Earnings Dashboard

Feature	Detail
Earnings View	Today / This Week / This Month — toggled by the rider
Per-Order Breakdown	Each completed order listed with commission earned
Rating	Current average star rating and total reviews
Safety Badge Status	'Safety Certified' badge shown — indicates training is current
Payout	Earnings processed daily or weekly via M-Pesa

7. Notification Architecture

Notifications keep the customer informed without requiring them to open the app. Every critical status change triggers both a push notification (for app users) and an SMS (for users without active data). This dual-channel approach is essential in the Kenyan market where data connectivity is not always guaranteed.

Trigger	Notification Message
Order placed	'Order #1042 received! Finding your rider...'
Rider assigned	'Brian is on his way with your gas'
Rider nearby (~5 mins)	'Almost there! Brian is 5 minutes away'
Delivered	'Delivered! Rate your experience'
10 min post-delivery	Safety tip push notification (see Section 8)
GasPoints milestone	'You've reached 500 points — Ksh 50 off your next order!'
Shop opens (7am)	Only sent to users who attempted to order when shop was closed
Referral converts	'Your friend just ordered! You've earned 250 GasPoints'
Referred friend's 3rd order	'Your friend placed their 3rd order! Bonus 100 GasPoints earned'

8. Safety Features

Safety is a powerful brand differentiator in the Kenyan LPG market, where gas explosions and counterfeit cylinders have caused public concern. These features build deep trust with customers and significantly reduce liability. Sections 8.1–8.2 are nice-to-have for the MVP; Section 8.3 (damaged cylinder flow) and 8.4 (SOS button) should be treated as MVP-required given the safety stakes.

8.1 Rider Safety Certification

Every rider is vetted, trained in LPG handling, and certified before their first delivery. Their certification status is displayed on the live tracking screen as a 'Safety Certified' badge alongside their name, photo, and rating. This is not just a feature — it is a trust signal that differentiates EldoGas from informal gas vendors.

8.2 Gas Leak Detection Tip — Auto Push Notification

Sent automatically 10 minutes after the rider marks an order as delivered. One notification, non-intrusive, and genuinely useful.

Sample Safety Push Notification

Safety Tip from EldoGas: Smell gas? Do not switch on lights or appliances. Open all windows, leave the building, and call 999 or 0800 723 723. Tap SOS below if you need immediate help.

8.3 One-Tap SOS Button

A permanent red shield icon sits in the top-right corner of the home screen — visible at all times but subtle enough not to alarm casual users. In a gas emergency, speed is critical. This button removes all friction.

SOS Button Flow

1. User taps the SOS shield icon
2. Confirmation: 'Are you experiencing a gas emergency?' → [YES, GET HELP] / [Cancel]
3. App shares live GPS location
4. Dials Kenya emergency services (999 / 0800 723 723)
5. Simultaneously sends SMS to shop: 'EMERGENCY — [Customer Name] at [Location Link] reported a gas emergency. Last order #1042'

Why this matters

No LPG delivery competitor in Kenya has a built-in gas emergency feature. This single button could save lives, protects EldoGas legally, and creates an enormous trust advantage in a market where gas safety anxiety is real and widespread.

8.4 Post-Delivery Rider Safety Checklist

Before the rider can mark an order as delivered, they must complete a 5-point checklist on their app. This creates accountability, ensures safe installation, and provides a digital audit trail for every delivery.

Checklist Item	Requirement
Cylinder Condition	Cylinder in good condition — no dents, rust, or damage
Valve Check	Valve secure, no gas smell detected
Safe Handover	Cylinder handed to customer safely and correctly
Empty Collected	Empty cylinder collected from customer (swap orders)
Payment Confirmed	Correct payment amount collected

9. GasPoints Loyalty Programme

GasPoints is a simple, transparent loyalty programme designed to deliver the first reward within 3–4 orders, creating habitual ordering behaviour from the very start. The earn and redeem rules are kept minimal, and the reward is always a concrete Kenyan shillings value that users immediately understand.

Design change from original concept

The original 1,000-point minimum threshold required 10 orders before any reward — too slow to build

habits. The revised structure introduces a 250-point first-order welcome bonus and a 500-point minimum redemption, bringing the first reward to within 3–4 orders for most customers.

9.1 Earning GasPoints

Action	Points Earned
First order — new customer welcome bonus	250 points
Complete a swap order	100 points
Complete a new cylinder order	150 points
Complete a 25kg or 50kg order	200 points
Leave a review	25 points
Refer a friend (friend places first order)	250 points
Referred friend completes their 3rd order	100 bonus points

9.2 Redeeming GasPoints

Redemption is automatic — the discount is applied at checkout when the customer chooses to redeem. The balance is always visible on the home screen, displayed like an M-Pesa balance so users check it habitually.

Points Balance	Redemption Value
500 points	Ksh 50 off delivery fee
1,000 points	Ksh 100 off delivery fee
2,000 points	Ksh 200 off any order (including gas cost)
5,000 points — Platinum status	Ksh 500 off any order + GasPoints Platinum badge on profile

9.3 Referral Flow

Referral Flow

1. Customer shares their unique referral code via WhatsApp, SMS, or social media
2. Friend downloads EldoGas and enters the code during onboarding
3. Friend places their first order
4. Referring customer receives push notification: 'Your friend just ordered! +250 GasPoints'
5. Points appear in referring customer's balance immediately
6. When the friend places their 3rd order, referring customer earns a bonus 100 GasPoints

9.4 GasPoints Milestone Notifications

Trigger	Notification Message
First-order bonus awarded	'Welcome! You've earned 250 GasPoints — you're 50% of the way to a Ksh 50 discount!'
Balance reaches 400 points	'Almost there! Just 100 more GasPoints for your first Ksh 50 reward.'
500-point threshold reached	'You've earned a Ksh 50 reward! Redeem it on your next order at checkout.'
1,000 points reached	'You've hit 1,000 GasPoints — that's Ksh 100 off your next order!'
Platinum milestone (5,000 pts)	'Congratulations! GasPoints Platinum status. Enjoy Ksh 500 off + your Platinum badge!'

10. Admin / Shop Dashboard

The shop owner needs full visibility and control from a web or mobile dashboard. This is the operational nerve centre — everything the rider apps and consumer app generate flows here. The Admin is also the sole authority for the product catalogue: all prices, sizes, brands, and add-ons are configured here.

10.1 Product Management Module

All product data is managed exclusively by the Admin. No price, size, or brand is hard-coded in the customer app. Changes take effect immediately across the live app.

Configuration Item	Description
Cylinder Sizes	Add, edit, or deactivate sizes. Deactivated sizes are hidden from the customer app instantly.
Gas Brands	Add or remove brands. Link each brand to the sizes it is available in.
Swap Refill Prices	Set the gas refill price per cylinder size. Updates live on save.
New Cylinder Prices	Set the cylinder hardware price and gas fill price separately per size.
Delivery Fees	Set a standard fee, or configure size-specific fees for commercial cylinders.
Add-On Items	Add, price, and link add-on accessories to specific cylinder sizes. Toggle active/inactive.
Add-On UI Type	Set the selection type per group: multi-select toggle (ADD/ADDED) or single-select radio.
Product Photos	Upload product photos for each add-on item to help customers identify accessories visually.
Price Change Log	Every price change is logged with a timestamp and the admin user who made the change.

10.2 Stock Management

Stock counts are a critical operational layer. The Admin sets and maintains filled cylinder counts per size; the system automatically deducts stock as riders pick up orders and alerts the Admin when levels are low.

Initial Stock Setup

Before go-live, the Admin sets the opening stock count for each active cylinder size under Product Management → Stock Counts, and sets a low-stock alert threshold per size. Any size with a count of 0 is automatically greyed out in the customer app.

Automatic Stock Deduction

Event	Stock Action	Triggered By
Customer places order	No change — order is queued	Customer app
Rider taps 'Picked Up'	Stock for that size reduced by 1	Rider app — automatic
Rider confirms delivery (swap)	Empty cylinder count increased by 1	Rider app — automatic
Rider reports out-of-stock	Admin alerted; order cancelled; stock stays at 0	Rider app
Admin manually adjusts stock	Count updated immediately in the live app	Admin dashboard

Low-Stock Alerts

Threshold	Alert Type	Message
At or below threshold (e.g., 5 units)	Dashboard badge + push notification	'Low stock: Only 5 × 13kg cylinders remaining. Consider restocking.'
Reaches 2 units	Urgent red alert in dashboard	'Critical: Only 2 × 13kg cylinders left.'
Reaches 0	Size auto-hidden from customer app	'13kg is out of stock and hidden from the app. Update stock to reactivate.'
Restored above 0	Size auto-reactivated in app	'13kg is back in stock. Customers can now order this size again.'

10.3 General Dashboard Features

Dashboard Feature	Function
Live Order Feed	All incoming orders in real time — manually assign or auto-assign to riders
Rider Status Board	Who is available, on a delivery, and where they are on the map
Stock Management	Current count of filled cylinders per size — see Section 10.2
Order History	Searchable and filterable — find any order by date, customer, or rider
Customer Database	Repeat customers, order history, address book
Revenue Summary	Daily and weekly totals broken down by payment method (Cash vs M-Pesa)
Rider Management	Add/remove riders, view stats, ratings, and certification status
Delivery Fee Settings	Set or adjust delivery fees per size — see Section 10.1
Notifications Control	Set shop open/closed status — triggers customer notifications

Dashboard Feature	Function
Stock Audit Log	Full log of every stock change: automatic deductions and manual adjustments

11. Exception & Edge-Case Flows

Every system that handles real-world deliveries must define how it behaves when things go wrong. The flows below give the development team clear, step-by-step handling for the most common failure scenarios. Each exception has a customer-facing resolution path, a rider-facing path, and an admin visibility requirement.

Design principle

Every exception should be resolved without the customer needing to call the shop. The app must surface the problem clearly, offer the customer a concrete next step, and alert the admin automatically. Phone calls to resolve simple errors destroy trust and increase churn.

11.1 Out of Stock — Requested Size Unavailable

Prevention first: if the Admin has marked a size as having zero stock, that size is greyed out in the customer app with the label 'Temporarily unavailable'. The customer cannot place an order for it.

If Stock Runs Out After an Order Is Placed

Out-of-Stock Exception Flow

1. Rider arrives at shop and finds the ordered size is out of stock.
2. Rider taps 'Report Issue' → selects 'Cylinder size out of stock'.
3. Customer receives push + SMS: 'We're sorry — we've run out of [size] today. Your order has been cancelled. No payment was taken.'
4. App shows customer two options: [Order a different size] or [Cancel — I'll order later].
5. If customer selects a different size, a new order is created at the correct price.
6. Admin dashboard flags the original order as 'Cancelled — Out of Stock'.
7. Admin receives alert: 'Stock for [size] has reached zero. Please restock before accepting further orders.'
8. Admin updates stock count once restocked; that size becomes orderable again automatically.

11.2 Wrong Cylinder Delivered

Discovered at the Door (Before Payment)

Wrong Cylinder — Discovered at Door

1. Customer notices the wrong cylinder before paying.
2. Customer taps 'There is a problem with my order' → selects 'Wrong size' or 'Wrong brand'.
3. Rider is notified: 'Customer reports an issue — please resolve before marking delivered.'
4. Rider confirms the error, returns to shop to collect the correct cylinder.
5. App updates status: 'Rider returning to shop — correction in progress.'
6. Customer receives: 'Your rider is collecting the correct cylinder. Updated ETA: ~X mins.'

7. Correct cylinder delivered, customer pays, order marked complete.
8. Admin sees the correction event logged in the order history.

Discovered After Delivery (Customer Has Paid)

Wrong Cylinder — Discovered After Delivery

1. Customer taps 'Report a delivery issue' → 'Wrong cylinder delivered'.
2. Admin receives an immediate alert: 'Post-delivery issue reported — Order #XXXX.'
3. Admin contacts the customer directly to arrange a correction delivery or refund.
4. Admin logs the resolution in the order record.
5. The rider's record is flagged with a 'wrong item' incident — tracked in rider performance stats.

11.3 Rider No-Show or Extreme Delay

Trigger	Action
Rider has not tapped 'Picked Up' within 15 mins of acceptance	Admin alert: 'Rider has not picked up Order #XXXX. Consider reassigning.'
Rider has not arrived within 2× the original ETA	Customer receives push + SMS: 'Your rider is delayed. Updated ETA coming shortly.'
Rider goes offline mid-delivery	Admin receives critical alert. Order flagged for reassignment.
Customer reports rider not arrived after ETA + 20 mins	Customer taps 'My rider hasn't arrived' — Admin alerted immediately.

Rider No-Show Resolution

1. Admin receives delay/no-show alert on dashboard.
2. Admin attempts to contact the rider via the dashboard call button.
3. If rider is unreachable, Admin taps 'Reassign Order' — dispatched to next available rider.
4. Customer receives: 'We're reassigning your delivery. Updated ETA: ~X mins. Sorry for the delay.'
5. If no rider available, Admin taps 'Cancel and Apologise' — customer receives cancellation notification. No payment taken.
6. The original rider's account is flagged with a 'missed delivery' incident for review.

11.4 Payment Dispute

Payment Dispute Flow

1. Rider taps 'Payment not confirmed' before marking delivered.
2. App shows rider: 'Ask the customer to show their M-Pesa confirmation SMS before handing over the cylinder.'

3. If customer shows the M-Pesa SMS, rider verifies the amount and till number match — marks paid.
4. If there is a genuine discrepancy, rider taps 'Escalate to shop' — Admin is alerted immediately.
5. Admin reviews the M-Pesa till transaction record and calls the customer if needed.
6. Admin resolves: either instructs rider to complete delivery or to return the cylinder.
7. All escalated payment disputes are logged in the order history with full notes.

Phase 2 — STK Push eliminates payment disputes

When M-Pesa STK Push is implemented (Phase 2), payment is initiated from within the app and confirmed automatically — eliminating manual payment disputes entirely. Prioritise for Phase 2.

11.5 Customer Cancellation

Order Stage at Cancellation	Allowed?	Consequence
Order placed — no rider assigned yet	Yes, free cancellation	Order cancelled instantly. No penalty. Admin notified.
Rider assigned — not yet picked up	Yes, with a warning	Customer sees a warning that rider is en route to pick up. Admin notified to stop rider.
Rider has picked up — en route	Restricted — Admin approval required	Customer must contact the shop. Admin can approve and redirect rider back.
Rider is at the door	Not permitted via app	Customer must resolve directly with the rider at the door.

11.6 Damaged or Unsafe Cylinder

Damaged Cylinder Flow

1. Customer taps 'Report a delivery issue' → 'Cylinder appears damaged or unsafe'.
2. App immediately shows safety guidance: 'Do not use the cylinder. Open windows, leave the area, and call us.'
3. App surfaces the one-tap SOS button and the shop phone number.
4. Admin receives an urgent alert: 'Damaged/unsafe cylinder reported — Order #XXXX — [Customer Location].'
5. Admin arranges an emergency replacement and logs the incident.
6. The damaged cylinder is tracked back to the specific stock item and removed from circulation.
7. The rider who delivered it receives a quality incident flag on their record.
8. Incident is documented in the admin dashboard for compliance and safety audit purposes.

This flow is safety-critical — treat as P0

Gas cylinder incidents have caused fatalities in Kenya. A clear, fast exception flow for damaged

cylinders is not just good UX — it is a legal and safety obligation. Include in the MVP.

12. MVP vs. Future Phases

The recommended approach is to launch with the core ordering and delivery experience fully polished, then layer in loyalty, safety, and intelligence features as the business scales.

Phase 1 — MVP (Launch With This)

Phase 1 — MVP

Consumer app: Onboarding, swap order flow, new cylinder flow (with add-ons for all sizes), live tracking,

pay on delivery, post-delivery rating, one-tap reorder

Rider app: Order notifications with accept/decline, active delivery flow, safety checklist, delivery confirmation

Admin dashboard: Order management, rider assignment, product catalogue management, stock management

(with low-stock alerts), revenue summary

Exception flows: All six exception scenarios in Section 11 (out of stock, wrong cylinder, rider no-show,

payment dispute, customer cancellation, damaged cylinder)

Safety: SOS button and damaged cylinder exception flow treated as P0 for the MVP

Phase 2 — After Early Traction

Phase 2

GasPoints loyalty programme and referral system

SOS emergency button (if not included in MVP)

Post-delivery safety push notification

Rider safety certification badge on live tracking screen

STK Push M-Pesa payment (eliminates payment disputes)

Saved multiple addresses (home, office, etc.)

Order history and one-tap reorder

Phase 3 — Scale

Phase 3

WhatsApp ordering chatbot (for customers without smartphones)

IoT gas level sensor integration for predictive ordering

Subscription and pre-paid refill plans

Referral programme dashboard with analytics

Expansion to second city (Mombasa, Kisumu, Nakuru)

13. Key Design & Operational Principles

Principle	Why It Matters
Delivery time is everything	Consistently under 45 minutes in urban areas. Under 30 in rush mode. This is the primary word-of-mouth driver.
SMS as a fallback	Every push notification is mirrored via SMS. Data access is not universal.
M-Pesa native	Till number payments are the default. STK Push (Phase 2) will eliminate manual payment steps.
Map pin over text address	Kenyan addresses are often informal. The GPS pin is the real address. Request location permission on onboarding.
Admin as source of truth	No price, size, or brand is hard-coded. The Admin dashboard controls the entire product catalogue in real time.
Safety as a brand	Safety Certified riders and the SOS button are not features — they are the brand promise.
Start dense, then expand	Launch in one urban area, prove the model, then expand. Do not spread stock and riders too thin early.
Exception flows are first-class	How the app handles failures defines customer trust more than how it handles success. Every error state must have a clear resolution path.

EldoGas — Gas delivered. No stress.

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